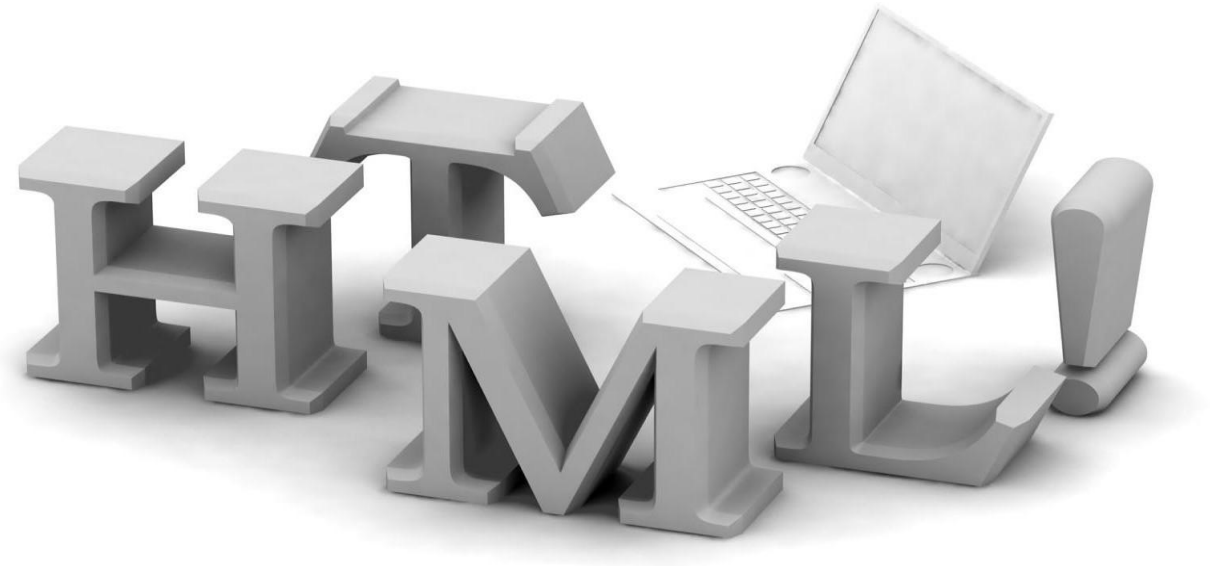
Faculty of Computer Science and Mathematics

Lab Exercise 2

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Matric Num:	S79184



Overview:

This in-class exercise extends Lab Exercise 2. You will apply tables (including colspan and rowspan), forms (text inputs, radio buttons, checkboxes, drop-down lists and labels) and semantic HTML tags (header, nav, section, article and footer) to solve a realistic problem.

There are two problems. Your lecturer will assign one problem to you, or you may be asked to complete both.

Learning Outcomes

1. Create tables with headers and merged cells using colspan and rowspan.

2. Build forms with appropriate input types, correctly linked labels and grouped radio buttons.

3. Structure a web page using semantic HTML tags.

4. Identify and correct errors in tables, forms and semantic structure, and validate code using the W3C Validator.

Best of luck!

Problem A: Faculty Event Page with Schedule and Registration

Your faculty is organising a one-day event, for example a Programming Workshop, a Career Day or a Hackathon. The organising committee needs a simple web page that shows the event schedule and allows students to register. Build the page using HTML only (no CSS). You may choose any event name, date and activities.

Instructions

Create a file named lab2_ExerA.html that contains the following:

1. Semantic Structure

- A `<header>` containing an `<h1>` with the event name and a short tagline.
- A `<nav>` with internal links (e.g. `href="#schedule"`, `href="#register"`, `href="#about"`) that jump to each section of the page.
- At least three `<section>` elements, each with an `id`: About the Event, Event Schedule and Registration.
- Inside the About section, two `<article>` elements, each introducing one speaker or one activity (with an `<h3>` and a `<p>`).
- A `<footer>` with the organiser name and a copyright line using `©`.

2. Event Schedule Table

- A table with `border="1"` and header cells (`<th>`) for Time, Activity, Venue and Person in Charge.
- At least 6 rows of activities from morning to evening.
- Use `colspan` at least once (e.g. a "Lunch Break" row that spans across Activity, Venue and Person in Charge).
- Use `rowspan` at least once (e.g. two consecutive sessions held in the same venue).

3. Registration Form

Create a form with `action="register.html"` and `method="post"` that contains:

- Text inputs for Full Name, Matric Number and E-mail, each with a `<label>` whose `for` matches the input `id`.
- Radio buttons for Year of Study (Year 1 – Year 4). All radio buttons must share the same `name` so only one can be selected.
- Checkboxes for the sessions the student wants to join (at least 3 choices).
- A drop-down list (`<select>`) for Programme or Faculty with at least 4 options.
- A submit button.

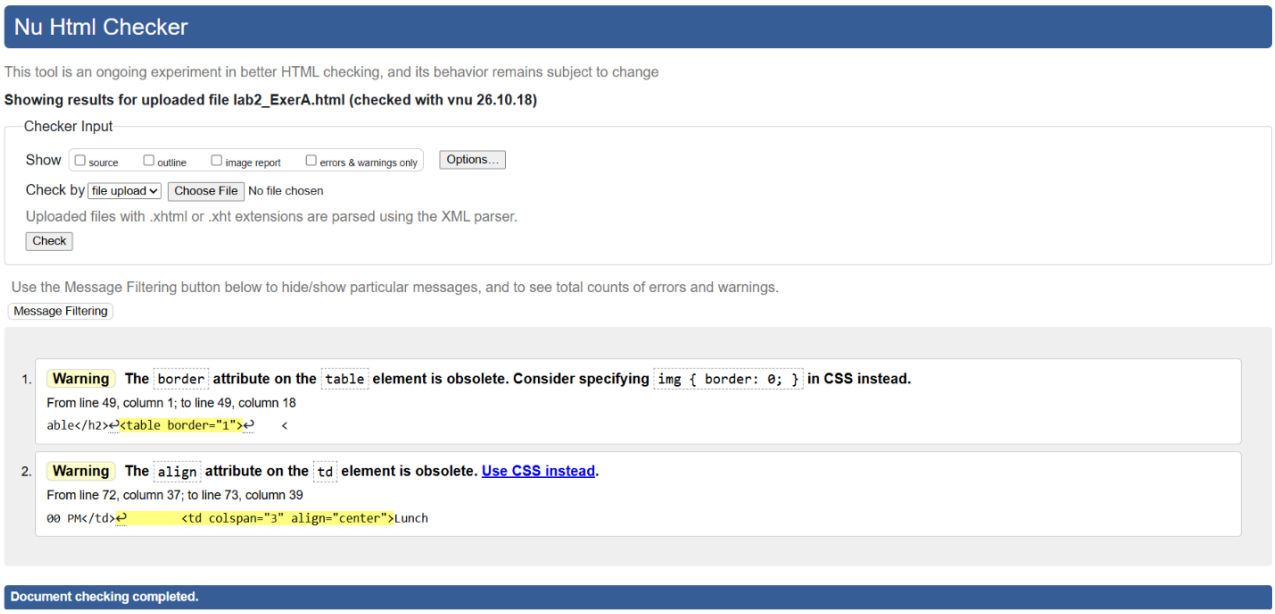
4. Comments

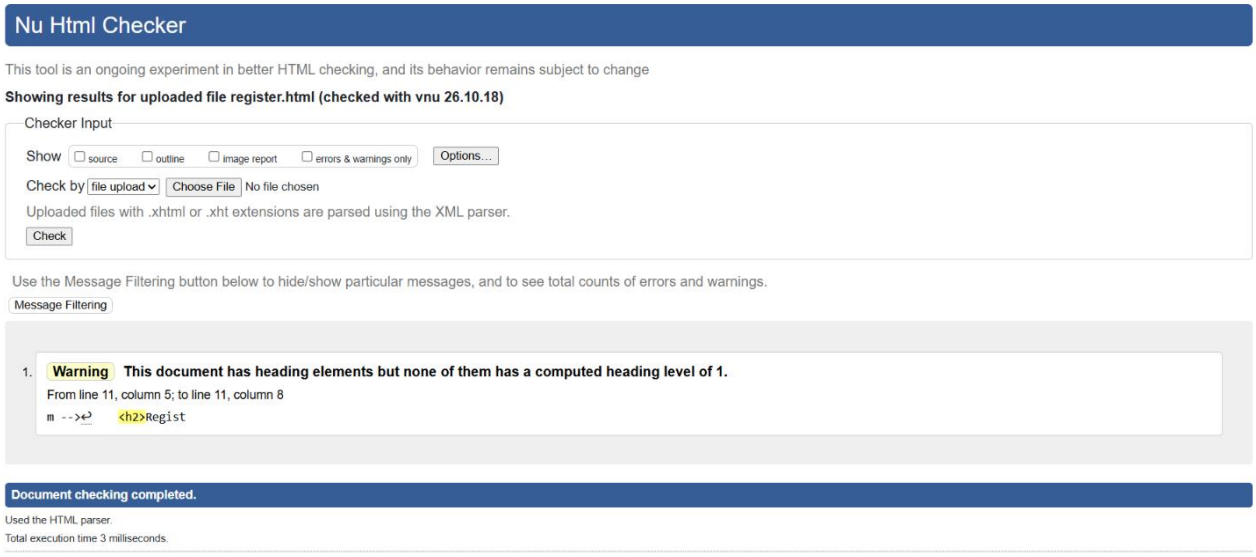
- Use comments (`<!-- ... -->`) to explain each section of your code.

Submission

Save the file as lab2_ExerA.html, validate it at `https://validator.w3.org` (Validate by File Upload), and submit it (Submit lab2_ExerA.html and Screenshot of validation) through Github

Validation:


Nu Html Checker
This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change
Showing results for uploaded file lab2_ExerA.html (checked with vnu 26.10.18)
Checker Input
Show source outline image report errors & warnings only Options…
Check by file upload Choose File No file chosen
Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.
Check
Use the Message Filtering button below to hide/show particular messages, and to see total counts of errors and warnings.
Message Filtering
1. Warning The border attribute on the table element is obsolete. Consider specifying img { border: 0; } in CSS instead.
From line 49, column 1; to line 49, column 18
able</h2>↩<table border="1">↩ <
2. Warning The align attribute on the td element is obsolete. Use CSS instead.
From line 72, column 37; to line 73, column 39
00 PM</td>↩ <td colspan="3" align="center">Lunch
Document checking completed.


Nu Html Checker
This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change
Showing results for uploaded file register.html (checked with vnu 26.10.18)
Checker Input
Show source outline image report errors & warnings only Options…
Check by file upload Choose File No file chosen
Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.
Check
Use the Message Filtering button below to hide/show particular messages, and to see total counts of errors and warnings.
Message Filtering
1. Warning This document has heading elements but none of them has a computed heading level of 1.
From line 11, column 5; to line 11, column 8
m -->↩ <h2>Regist
Document checking completed.
Used the HTML parser.
Total execution time 3 milliseconds.

Code(lab2_ExerA):

lab2_ExerA.html > html > head > table > tr > td

```
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>Programming Workshop</title>

<!--Header: Event Name and Tagline-->
    <header>
        <h1>Programming Workshop</h1>
        <p>Learn, Code, and Create!</p>

    </header>
<!--Navigation Menu: Links to Schedule, Register, and About sections-->
    <nav>
        <a href="#schedule">Schedule</a> |
        <a href="#register">Register</a>  | <a href="#about">About</a>
    </nav>
<!--About the event -->
    <section id="about">
    <h2>About the Event</h2>
    <p>
        Programming Workshop is an event designed to help
        students improve their programming skills through
        practical exercises and coding activities.
    </p>
<!--Introducing the speaker -->
    <article>
        <h3>Dr. Muhamad Arif </h3>
        <p>
        Dr. Muhamad Arif is an experienced software developer
        who will share his knowledge of programming
        fundamentals, coding techniques and problem-solving
        skills with students.
        </p>
    </article>
```

```
<!--Activity Description -->
    <article>
        <h3>Hands-on Coding Session</h3>
        <p>
            Students will practise writing code and solving
            programming problems.
        </p>
    </article>
</section>

<!--Schedule Table: Time, Activity, Venue, and Person in Charge-->
<h2>Event Schedule Table</h2>
<table border="1">
    <tr>
        <th>Time</th>
        <th>Activity</th>
        <th>Venue</th>
        <th>Person in Charge</th>
    </tr>

    <tr>
        <td>9:00 AM - 10:00 AM</td>
        <td>Introduction to Programming</td>
        <td>Room 101</td>
        <td>Dr. Muhamad Arif</td>
    </tr>

    <tr>
        <td>10:15 AM - 11:15 AM</td>
        <td>Hands-on Coding Session</td>
        <td>Room 102</td>
        <td>Dr. Muhamad Arif</td>
    </tr>
```

```
    <tr>
        <td>11:15 AM - 12:00 PM</td>
        <td colspan="3" align="center">Lunch Break</td>
    </tr>

    <tr>
        <td>12:30 PM - 1:30 PM</td>
        <td rowspan="2">Problem Solving Techniques</td>
        <td>Room 103</td>
        <td>Dr. Muhamad Arif</td>
    </tr>

    <tr>
        <td>1:45 PM - 2:45 PM</td>
        <td>Room 103</td>
        <td>Dr. Muhamad Arif</td>
    </tr>

    <tr>
        <td>3:00 PM - 4:00 PM</td>
        <td>Q&A Session</td>
        <td>Room 105</td>
        <td>Dr. Muhamad Arif</td>
    </tr>

</table>

<!-- Footer: Organiser name and copyright -->
<footer>
    <p>Organised by the Faculty of Computer Science and Mathematics</p>
    <p>&copy; 2026 Programming Workshop. All rights reserved.</p>
</footer>
```

Code(register)

```
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>Registration Form</title>
    </head>
    <body>

<!-- Registration Form -->
    <h2>Registration Form</h2>
    <form action="register.html" method="post">
        <label for="name">Full Name:</label>
        <input type="text" id="name" name="name" required><br><br>

        <label for="Mnumber">Matric Number:</label>
        <input type="tel" id="Mnumber" name="Mnumber"><br><br>

        <label for="email">Email:</label>
        <input type="email" id="email" name="email" required><br><br>

        <label for="year1">Year of Study:</label><br>
        <input type="radio" id="year1" name="yearOfStudy" value="1" required>
        <label for="year1">1</label><br>
        <input type="radio" id="year2" name="yearOfStudy" value="2" required>
        <label for="year2">2</label><br>
        <input type="radio" id="year3" name="yearOfStudy" value="3" required>
        <label for="year3">3</label><br>
        <input type="radio" id="year4" name="yearOfStudy" value="4" required>
        <label for="year4">4</label><br><br>

        <label for="session">Program:</label><br>
        <select id="session" name="session">
            <option value="Software">Software Engineering</option>
            <option value="Computer">Computer Science</option>
            <option value="Information">Information Technology</option>
```

```
        </select><br><br>

        <input type="submit" value="Register">
    </form>
</body>
```

Problem B: Course Feedback Page (Debugging and Extend)

A student helper created a Course Feedback page for CSF3133. The page shows a feedback summary table and a feedback form, but it contains several errors. Some radio buttons can be selected at the same time, clicking some labels does nothing, and the table looks broken. Your tasks are to find and fix the errors, then extend the page.

Given Code

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8">
    <title>Course Feedback</title>
</head>
<body>
<header>
    <h1>CSF3133 Course Feedback</h1>
</head>
<nav>
    <a href="#results">Results</a> | <a href="#form">Feedback Form</a>
</nav>
<section id="results">
    <h2>Feedback Summary</h2>
    <table border="1">
        <tr>
            <th>Week</th>
            <th>Topic</th>
            <td>Rating</td>
        </tr>
        <tr>
            <td>1</td>
            <td>Introduction to HTML</td>
            <td>4.5</td>
        <tr>
            <td>2</td>
            <td colspan="3">Tables and Forms</td>
        </tr>
        <tr>
            <td>3</td>
            <td>Semantic HTML<td>4.0</td>
        </tr>
    </table>
</section>
<section id="form">
    <h2>Give Your Feedback</h2>
    <form action="submit_feedback.php" method="post">
        <label for="name">Name:</label>
        <input type="text" id="fullname" name="name"><br>
        <label>Overall rating:</label><br>
        <input type="radio" id="good" name="good" value="good">
        <label for="good">Good</label>
        <input type="radio" id="poor" name="poor" value="poor">
        <label for="poor">Poor</label><br>
        <input type="checkbox" id="lab" name="lab" value="yes">
        <label for="labs">I attended all labs</label><br>
        <input type="submit" value="Submit">
    </from>
</section>
<footer>
    <p>Copyright &copy 2026 CSF3133</p>
</footer>
</body>
</html>
```

Instructions

Part 1. Find and Fix the Errors

1. Copy the given code into a new file named lab2_ExerB.html.

2. Identify and correct all errors. At the top of the `<body>`, write an HTML comment listing each error and the line number, e.g. `<!-- Error 1 (Line 1): missing <!DOCTYPE html> -->`.

4. Validate your corrected file at `https://validator.w3.org`. Your final page must show "No errors or warnings to show".

Part 2. Extend into a Two-Page Website

5. Table: add rows for Week 4 and Week 5. Both weeks cover the same topic "Mini Project", so use `rowspan="2"` on the Topic cell. Add a final row "Average Rating" that uses `colspan="2".`

6. Form: add an E-mail text input with a matching `<label>`, a third radio option "Average", a group of at least 3 checkboxes "Topics I want more practice on", and a `<select>` for Section/Group (at least 3 options).

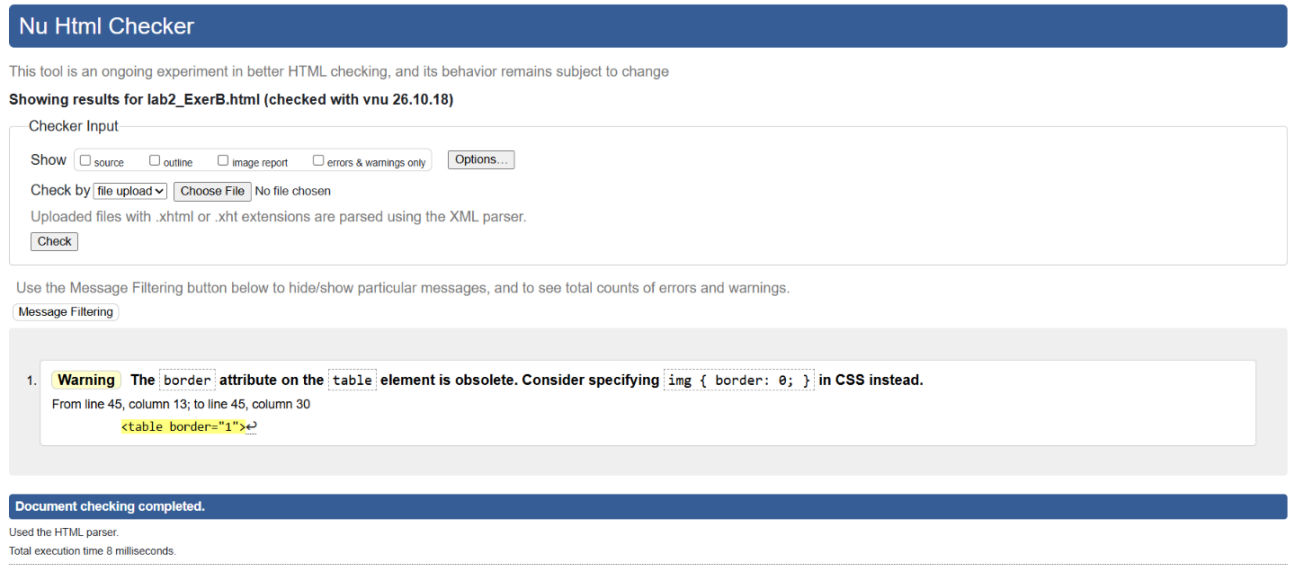
7. Semantic HTML: add a new `<section id="comments">` titled "Student Comments" that contains two <article> elements, each with an `<h3>` (student name) and a `<p>` (comment). Add a link to this section in the `<nav>`.

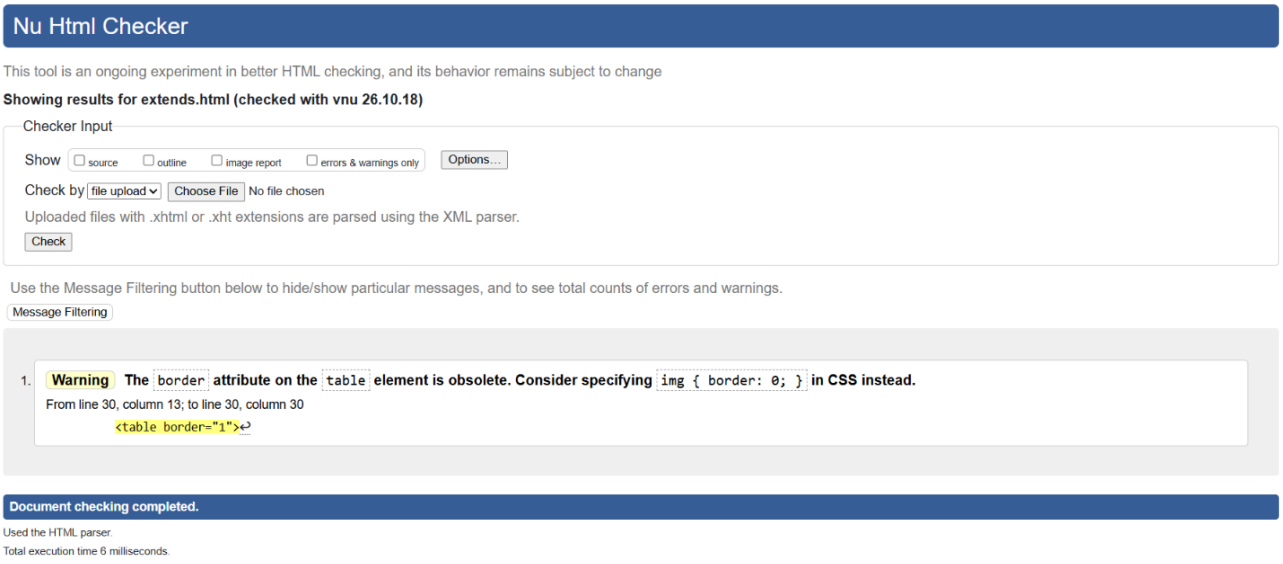
8. Use comments to explain each new section of your code.

Submission

Save the file as lab2_ExerB.html, validate it at `https://validator.w3.org` (Validate by File Upload), and submit it (Submit lab2_ExerB.html and Screenshot of validation) through Github

Validation:


Nu Html Checker
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Checker Input
Show source outline image report errors & warnings only Options…
Check by file upload Choose File No file chosen
Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.
Check
Use the Message Filtering button below to hide/show particular messages, and to see total counts of errors and warnings.
Message Filtering
1. Warning The border attribute on the table element is obsolete. Consider specifying img { border: 0; } in CSS instead.
From line 45, column 13; to line 45, column 30
<table border="1">
Document checking completed.
Used the HTML parser.
Total execution time 8 milliseconds.


Nu Html Checker
This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change
Showing results for extends.html (checked with vnu 26.10.18)
Checker Input
Show source outline image report errors & warnings only Options…
Check by file upload Choose File No file chosen
Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.
Check
Use the Message Filtering button below to hide/show particular messages, and to see total counts of errors and warnings.
Message Filtering
1. Warning The border attribute on the table element is obsolete. Consider specifying img { border: 0; } in CSS instead.
From line 30, column 13; to line 30, column 30
<table border="1">
Document checking completed.
Used the HTML parser.
Total execution time 6 milliseconds.

Code(lab2_ExerB)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Course Feedback</title>
</head>

<body>

    <!--
    Problem B: Errors found

    1. Line 10: Incorrect closing tag </head> instead of </header>.
    2. Line 20: Used <td> instead of <th> for the Rating heading.
    3. Line 26: Missing closing </tr> tag in the Week 1 row.
    4. Line 28: Incorrect colspan="3" for Tables and Forms.
    5. Line 32: Missing closing </td> after Semantic HTML.
    6. Line 40: Label for="name" did not match input id="fullname".
    7. Lines 42 and 44: Radio buttons used different name attributes.
    8. Line 47: Label for="labs" did not match checkbox id="lab".
    9. Line 49: Incorrect closing tag </from> instead of </form>.
    10. Line 52: Missing semicolon in the &copy; entity.
    -->

    <!-- Header -->
    <header>
        <h1>CSF3133 Course Feedback</h1>
    </header>

    <!-- Navigation -->
    <nav>
        <a href="#results">Results</a> |
        <a href="#form">Feedback Form</a> |
        <a href="#comments">Student Comments</a>
    </nav>
```

```
    <main>

        <!-- Course feedback summary table -->
        <section id="results">
            <h2>Feedback Summary</h2>

            <table border="1">
                <thead>
                    <tr>
                        <th>Week</th>
                        <th>Topic</th>
                        <th>Rating</th>
                    </tr>
                </thead>

                <tbody>
                    <tr>
                        <td>1</td>
                        <td>Introduction to HTML</td>
                        <td>4.5</td>
                    </tr>

                    <tr>
                        <td>2</td>
                        <td>Tables and Forms</td>
                        <td>4.0</td>
                    </tr>

                    <tr>
                        <td>3</td>
                        <td>Semantic HTML</td>
                        <td>4.0</td>
                    </tr>
```

```
                    <tr>
                        <td>4</td>
                        <td rowspan="2">Mini Project</td>
                        <td>4.5</td>
                    </tr>

                    <tr>
                        <td>5</td>
                        <td>5.0</td>
                    </tr>

                    <tr>
                        <td colspan="2">Average Rating</td>
                        <td>4.4</td>
                    </tr>
                </tbody>
            </table>
        </section>

        <!-- Feedback form -->
        <section id="form">
            <h2>Give Your Feedback</h2>

            <form action="submit_feedback.php" method="post">

                <!-- Name -->
                <p>
                    <label for="fullname">Name:</label>
                    <input type="text" id="fullname"
                           name="fullname" required>
                </p>
```

```
105             <!-- Email -->
106             <p>
107                 <label for="email">Email:</label>
108                 <input type="email" id="email"
109                        name="email" required>
110             </p>
111
112             <!-- Overall rating -->
113             <fieldset>
114                 <legend>Overall Rating:</legend>
115
116                 <input type="radio" id="good"
117                        name="rating" value="good" required>
118                 <label for="good">Good</label>
119
120                 <input type="radio" id="average"
121                        name="rating" value="average">
122                 <label for="average">Average</label>
123
124                 <input type="radio" id="poor"
125                        name="rating" value="poor">
126                 <label for="poor">Poor</label>
127             </fieldset>
128
129             <br>
130
131             <!-- Topics to practise -->
132             <fieldset>
133                 <legend>Topics to Practise More:</legend>
134
135                 <input type="checkbox" id="html"
```

```
136                        name="topics" value="html">
137                 <label for="html">HTML</label>
138
139                 <input type="checkbox" id="tables"
140                        name="topics" value="tables">
141                 <label for="tables">Tables</label>
142
143                 <input type="checkbox" id="forms"
144                        name="topics" value="forms">
145                 <label for="forms">Forms</label>
146             </fieldset>
147
148             <br>
149
150             <!-- Section or group selection -->
151             <p>
152                 <label for="group">Section/Group:</label>
153                 <select id="group" name="group" required>
154                     <option value="">-- Select Group --</option>
155                     <option value="A">Group A</option>
156                     <option value="B">Group B</option>
157                     <option value="C">Group C</option>
158                 </select>
159             </p>
160
161             <!-- Additional feedback -->
162             <p>
163                 <label for="feedback">Additional Comments:</label>
164                 <br>
165                 <textarea id="feedback" name="feedback"
166                           rows="4" cols="40"></textarea>
```

```
            </p>

            <input type="submit" value="Submit">
            <input type="reset" value="Clear">

        </form>
    </section>

    <!-- Student comments -->
    <section id="comments">
        <h2>Student Comments</h2>

        <article>
            <h3>Student 1</h3>
            <p>
                This course helped me understand HTML structure
                and how to create a basic webpage.
            </p>
        </article>

        <article>
            <h3>Student 2</h3>
            <p>
                I enjoyed learning about tables and forms.
                More practical exercises would be helpful.
            </p>
        </article>

        <p><a href="#top">Return to Top</a></p>
    </section>

```

```
</main>

<!-- Footer -->
<footer>
    <p>Copyright &copy; 2026 CSF3133</p>
</footer>

</body>
</html>
```

Code(extend):

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Course Feedback</title>
</head>

<body id="top">
    <!-- Header -->
    <header>
        <h1>CSF3133 Course Feedback</h1>
        <p>Share your feedback about this course.</p>
    </header>

    <!-- Navi -->
    <nav>
        <a href="#feedback">Course Feedback</a> |
        <a href="#form">Feedback Form</a> |
        <a href="#comments">Student Comments</a>
    </nav>

    <main>

        <!-- Section 1: Course feedback table -->
        <section id="feedback">
            <h2>Weekly Course Feedback</h2>

            <table border="1">
                <thead>
                    <tr>
                        <th>Week</th>
                        <th>Topic</th>
                        <th>Rating</th>
                    </tr>
                </thead>
```

```
                <tbody>
                    <tr>
                        <td>Week 1</td>
                        <td>Introduction to HTML</td>
                        <td>Good</td>
                    </tr>

                    <tr>
                        <td>Week 2</td>
                        <td>Tables and Forms</td>
                        <td>Good</td>
                    </tr>

                    <tr>
                        <td>Week 3</td>
                        <td>Semantic HTML</td>
                        <td>Good</td>
                    </tr>

                    <tr>
                        <td>Week 4</td>
                        <td rowspan="2">Mini Project</td>
                        <td>Good</td>
                    </tr>

                    <tr>
                        <td>Week 5</td>
                        <td>Good</td>
                    </tr>

                    <tr>
```

```
                    <td colspan="2">Average Rating</td>
                    <td>Good</td>
                </tr>
            </tbody>
        </table>
    </section>

    <!-- Section 2: Feedback form -->
    <section id="form">
        <h2>Course Feedback Form</h2>

        <form action="#" method="post">

            <!-- Student name -->
            <p>
                <label for="fullname">Full Name:</label>
                <input type="text"
                       id="fullname"
                       name="fullname"
                       required>
            </p>

            <!-- Student email -->
            <p>
                <label for="email">Email:</label>
                <input type="email"
                       id="email"
                       name="email"
                       required>
            </p>

            <!-- Course rating radio buttons -->
            <fieldset>
                <legend>How would you rate this course?</legend>

                <label>
                    <input type="radio"
                           id="good"
                           name="rating"
                           value="Good"
                           required>
                    Good
                </label>

                <label>
                    <input type="radio"
                           id="average"
                           name="rating"
                           value="Average">
                    Average
                </label>

                <label>
                    <input type="radio"
                           id="poor"
                           name="rating"
                           value="Poor">
                    Poor
                </label>
            </fieldset>

            <br>
```

```
133             <!-- Topics to practise checkboxes -->
134             <fieldset>
135                 <legend>Which topics would you like to practise more?</legend>
136 
137                 <label>
138                     <input type="checkbox"
139                            id="html"
140                            name="topics"
141                            value="HTML">
142                     HTML
143                 </label>
144 
145                 <label>
146                     <input type="checkbox"
147                            id="tables"
148                            name="topics"
149                            value="Tables">
150                     Tables
151                 </label>
152 
153                 <label>
154                     <input type="checkbox"
155                            id="lab"
156                            name="topics"
157                            value="Forms">
158                     Forms
159                 </label>
160             </fieldset>
161 
162             <br>
163 
```

```
164             <!-- Section or group selection -->
165             <p>
166                 <label for="group">Section/Group:</label>
167                 <select id="group" name="group" required>
168                     <option value="">-- Select Group --</option>
169                     <option value="A">K1</option>
170                     <option value="B">K2</option>
171                     <option value="C">K3</option>
172                 </select>
173             </p>
174 
175             <!-- Additional comments -->
176             <p>
177                 <label for="feedback-comments">
178                     Additional Feedback:
179                 </label>
180                 <br>
181                 <textarea id="feedback-comments"
182                           name="feedback-comments"
183                           rows="4"
184                           cols="40"
185                           placeholder="Write your feedback here..."></textarea>
186             </p>
187 
188             <!-- Form buttons -->
189             <button type="submit">Submit Feedback</button>
190             <button type="reset">Clear Form</button>
191 
192         </form>
193 
```

```
194             <p><a href="#top">Return to Top</a></p>
195         </section>
196
197         <!-- Section 3: Student comments -->
198         <section id="comments">
199             <h2>Student Comments</h2>
200
201             <article>
202                 <h3>Muhammad Raziq (S76345)</h3>
203                 <p>
204                     The course helps me understand the basic structure
205                     of HTML and how to create a webpage.
206                 </p>
207             </article>
208
209             <article>
210                 <h3>Muhammad Azeem Syahmi (S76856)</h3>
211                 <p>
212                     I enjoy learning about tables and forms.
213                     More practical exercises would help me improve my skills.
214                 </p>
215             </article>
216
217             <p><a href="#top">Return to Top</a></p>
218         </section>
219
220     </main>
221
222     <!-- Footer -->
223     <footer>
224         <p>&copy; 2026 Course Feedback Page</p>
225     </footer>
```

Rubric

Table A: Rubric for Additional Lab Exercise 2 – Problem A

Criteria	Description	Marks
Table Creation	Schedule table has <th> headers, at least 6 rows of meaningful data, and correct use of colspan and rowspan (no extra or missing cells).	3
Form Structure	Form contains text inputs with matching <label for> and id, a grouped set of radio buttons (same name), at least 3 checkboxes, a <select> with at least 4 options, and a submit button.	3
Semantic HTML	Proper use of <header>, <nav> with working internal links, <section> (min. 3), <article> (min. 2) and <footer>.	2
Code Organization and Readability	Code is well-organized, properly indented and commented. File is named lab2a.html and passes the W3C Validator.	2
Total		**10**

Table B: Rubric for Additional Lab Exercise 2 – Problem B

Criteria	Description	Marks
Error Identification	At least 7 errors correctly identified with line numbers and listed in an HTML comment.	3
Error Correction & Validation	All errors corrected. Radio buttons work as one group, all labels are linked, and the page passes the W3C Validator.	2
Table Extension	Week 4 and Week 5 rows added with correct rowspan; Average Rating row uses correct colspan. No extra or missing cells.	2
Form Extension	E-mail input with label, third radio option in the same group, at least 3 checkboxes and a <select> with at least 3 options.	2
Semantic HTML and Code Organization	New <section> with two <article> elements and a working <nav> link. Code is indented and commented.	1
Total		**10**